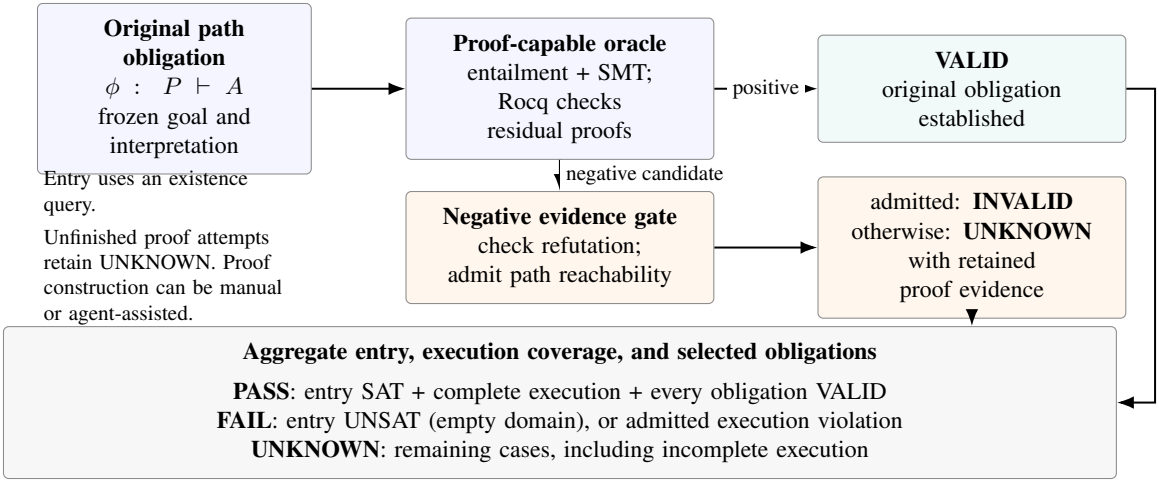

Original path obligation
$\phi : P \vdash A$
frozen goal and interpretation
Entry uses an existence query.
Unfinished proof attempts retain UNKNOWN. Proof construction can be manual or agent-assisted.
Proof-capable oracle
entailment + SMT; Rocq checks residual proofs
positive
VALID
original obligation established
negative candidate
Negative evidence gate
check refutation; admit path reachability
admitted: INVALID
otherwise: UNKNOWN
with retained proof evidence
Aggregate entry, execution coverage, and selected obligations
PASS: entry SAT + complete execution + every obligation VALID
FAIL: entry UNSAT (empty domain), or admitted execution violation
UNKNOWN: remaining cases, including incomplete execution